# Advanced Topics and Trends in Blockchain

Layer-2 Scaling | DeFi | NFTs

# Ethereum Layer 2(L2)

- L2 solutions are secondary protocols built on top of Ethereum (Layer 1) to enhance scalability, providing faster transactions and lower fees while inheriting Ethereum's security.
- L2 processing transactions off-chain and posting aggregated data back to Ethereum, significantly reducing congestion.
- Common types include Optimistic Rollups and ZK-Rollups.

# Blockchain Trilemma

- it’s hard for blockchains to achieve optimal levels of decentralization, security, and scalability simultaneously.

# Etherium Layer 1


- The Ethereum network's resources, or layer 1 scaling solutions, is the original blockchain that runs on the Ethereum consensus protocol for data processing tasks.
- The existing architecture of blockchain layer networks of nodes that validate transactions and execute smart contracts and help with scaling solutions and data processing load state channels.

- To process data simultaneously, Ethereum will automate transactions on the underlying distributed ledger of the main blockchain network. The smart contract with the terms of the agreement between buyer and seller is directly written into lines of code.

- This network processes the automatic execution of the contract without the need for intermediaries, which can save time and reduce the risk of fraud.

# Layer 1 High Gas Fees Collapse Of CryptoKitties

- In November 2017, the first big blockchain layer game CryptoKitties flooded the Ethereum network with users and super high gas fees. However, there was a problem with layer 1. The flood of new users created a bad user experience and bad scaling solutions.

# What's a CryptoKitty?

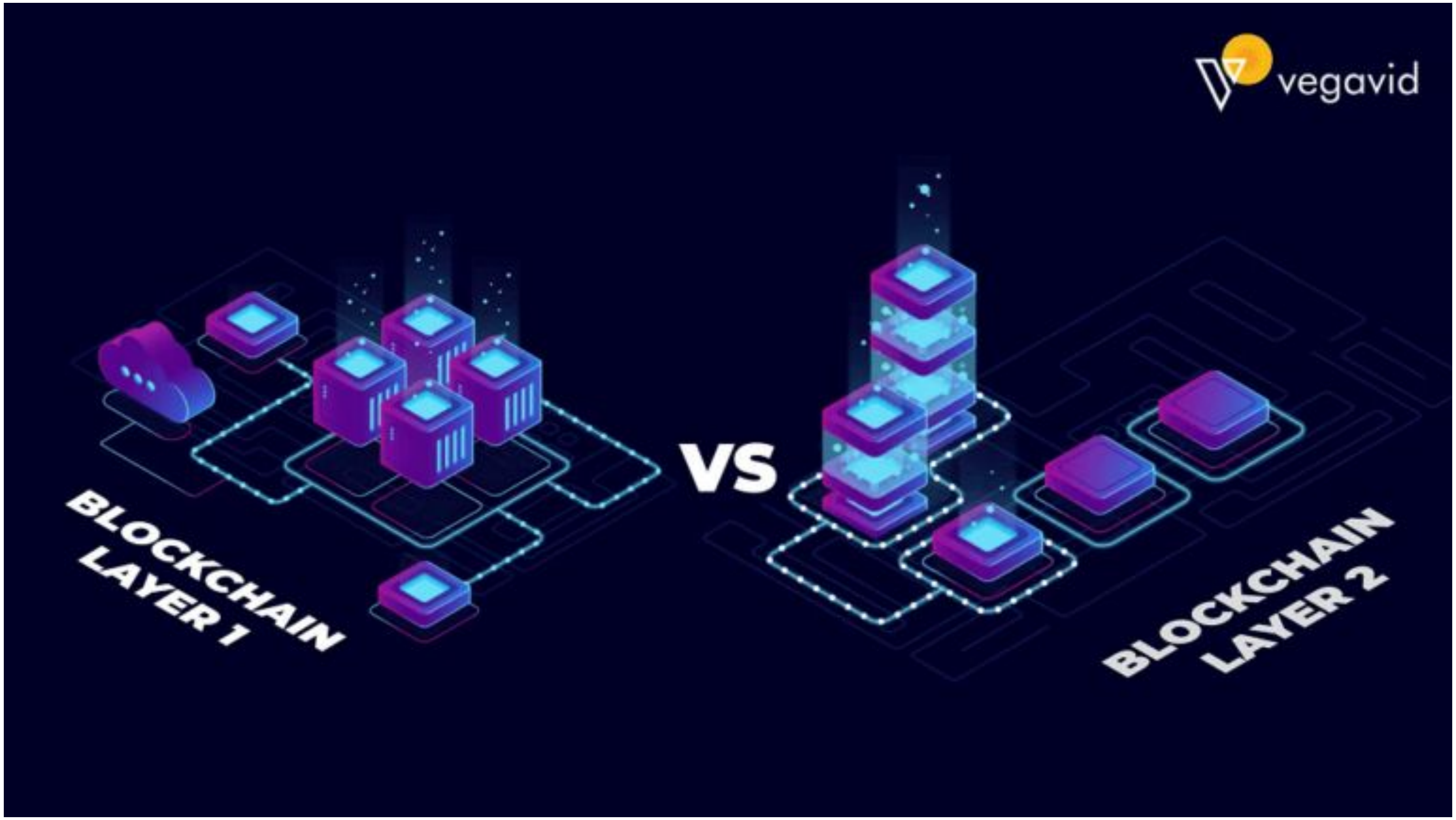
**Each CryptoKitty is a token, a set of data on the Ethereum blockchain. Unlike the cryptocurrencies Ethereum and Bitcoin, these tokens are nonfungible; that is, they are not interchangeable.**

| Unique ID | Mother's ID, father's ID | Genes |
|---|---|---|
| The unique ID makes a CryptoKitty a nonfungible token. | The token contains the kitty's lineage and other data. | The kitty's genes determine its unique look. |

DAPPER LABS

# Layer 2: Improved Blockchain Scalability

- The consensus protocol improvements for the scaling solution is Layer 2 of Ethereum and refers to additional consensus protocols and technologies built on top of the Ethereum layer 1 blockchain.
- These transaction requests protocols and technologies are designed to improve the scalability, speed, and efficiency of the Ethereum network by offloading some of the workloads from the main blockchain decentralized system.
- layer 2 technologies on Ethereum include Arbitrum One, Optimism, and Boba Network.


vegavid
BLOCKCHAIN
LAYER 1
VS
BLOCKCHAIN
LAYER 2

# Layer 2 Scaling Solutions

- Polygon
- Lightning Network
- Optimistic Rollups

# Optimistic Rollups

- Fraud proof
- Batch transactions
- Lower cost

# Optimistic Rollups

- Processing transactions off of the main Ethereum chain (Layer 1) and only settling final states
- Optimistic rollups work by aggregating or "batching" numerous transactions off-chain and then submitting them to the Ethereum network as a single transaction.
- Aggregation takes place within a smart contract on the Ethereum blockchain, known as the rollup contract.
- The essential components of this process are the state roots and Merkle roots, which serve as cryptographic proofs of the state of transactions within the Rollup.

# Optimistic Rollups

- **State Roots:** These are cryptographic hashes that represent the state of the rollup at a given time. Each batch of transactions submitted to the Ethereum mainnet includes a pre-state root (representing the state before the batch of transactions) and a post-state root (representing the state after the transactions). This allows the Ethereum network to verify the integrity of the transactions processed by the rollup.

- **Merkle Roots:** The Merkle root is a single hash that summarizes all the transactions in a batch. It enables anyone to verify the inclusion of a specific transaction in the batch by providing a Merkle proof without having to reveal the entire set of transactions. This feature is critical for maintaining privacy and efficiency in transaction processing.

# Sequencers and Data Availability

- A sequencer is a node within the optimistic rollup network responsible for aggregating transactions, generating state roots, and submitting them to the Ethereum blockchain.
- The sequencer plays a crucial role in ensuring data availability. It does so by posting the data associated with each batch of transactions (e.g., calldata) on-chain, allowing other nodes to reconstruct the state of the rollup if needed.
- This mechanism ensures the security and integrity of the rollup, even if the sequencer behaves maliciously or becomes unavailable.

# Fraud Proofs and Dispute Resolution

- One might wonder: if transactions are processed off-chain, how can we ensure their validity? This is where the concept of fraud proofs and dispute resolution comes into play.
- The dispute resolution mechanism in optimistic rollups is a critical component designed to ensure the integrity and security of transactions processed off the Ethereum mainnet. This mechanism allows any participant in the network to challenge the validity of transactions within a specific period after they have been posted to the Ethereum blockchain

- **1. The Challenge Window**
- After a batch of transactions has been submitted to the Ethereum mainnet by the sequencer, there's a predefined challenge period or window. This period is crucial as it provides time for network participants to review the batched transactions for any discrepancies or fraudulent activities. The typical duration of this challenge window can vary but is often set to a week or more to allow ample time for thorough review.

- **2. Submitting a Fraud Proof**
- If, during the challenge period, a participant (referred to as a "verifier") detects a fraudulent transaction within the batch, they can submit a fraud proof to the Ethereum mainnet. This fraud proof is essentially evidence that a particular transaction within the batch is invalid. The proof requires demonstrating the incorrectness of the transaction without revealing private data related to other transactions in the batch. This is often achieved through cryptographic proofs that leverage the unique properties of Merkle trees and state roots.

- **3. The Dispute Resolution Process**

Upon submission of a fraud proof, the optimistic rollup's smart contract on the Ethereum blockchain initiates a dispute resolution process. This process involves:

- **Verification of the Fraud Proof:** The smart contract examines the submitted fraud proof to determine its validity. This may involve recomputing the state transition or executing the disputed transaction in a sandboxed environment within the smart contract to verify if it leads to the same state transition as claimed by the sequencer.
- **Bisection and Iteration:** In more complex disputes, the process may involve a bisection method. In this method, the disputed batch is split into smaller parts, and the challenger is asked to specify which part contains the fraud. This iterative process narrows down the disputed transaction(s) to a manageable size, making it easier to verify the fraud claim.
- **Resolution:** If the fraud proof is found to be valid, the fraudulent transaction is nullified, and the rollup is reverted to its correct state before the fraudulent transaction. The sequencer who submitted the fraudulent batch may be penalized, often by forfeiting a bond or deposit. If the fraud proof is invalid, the transactions are finalized, and the challenger may face a penalty for the false accusation.

- **4. Finality and Security**
- This dispute resolution mechanism ensures that all transactions within an optimistic rollup achieve finality only after the challenge window has closed without any successful fraud proofs. It underscores the security of optimistic rollups by ensuring that only valid transactions are finalized and incorporated into the Ethereum blockchain. Moreover, it reinforces the principle that the security of the network does not rely solely on the honesty of the sequencer but is upheld by the vigilance of the entire network and the ability to challenge and verify transactions.
- In essence, the dispute resolution mechanism in optimistic rollups is designed to leverage the collective security and verification capabilities of the Ethereum network, ensuring that despite operating off-chain for scalability, the integrity and trustlessness of the blockchain are maintained.

# Transaction finality

- Transaction finality in blockchain technology is the assurance that once a transaction has been confirmed, it cannot be reversed or altered.

# Advantages of Optimistic Rollups

- The advantages of optimistic rollups are multifold. They drastically reduce transaction fees and increase throughput, making Ethereum more scalable and efficient. Their compatibility with the Ethereum Virtual Machine (EVM) ensures that existing dApps can easily migrate to optimistic rollups, leveraging the enhanced performance without significant changes to their codebase. Moreover, the optimistic execution model fosters a secure and trustworthy environment, which is crucial for the growth and adoption of decentralized technologies.

# Challenges and Limitations of Optimistic Rollups

- Despite their potential, optimistic rollups are not without challenges. The dispute resolution period, necessary for fraud detection, introduces a delay in withdrawing funds from Layer 2 to Layer 1, potentially affecting liquidity and user experience. Furthermore, the reliance on sequencers raises concerns about centralization and data availability. However, ongoing improvements and innovations within the ecosystem are addressing these issues, pushing optimistic rollups closer to their ideal state of efficiency, security, and user-friendliness.

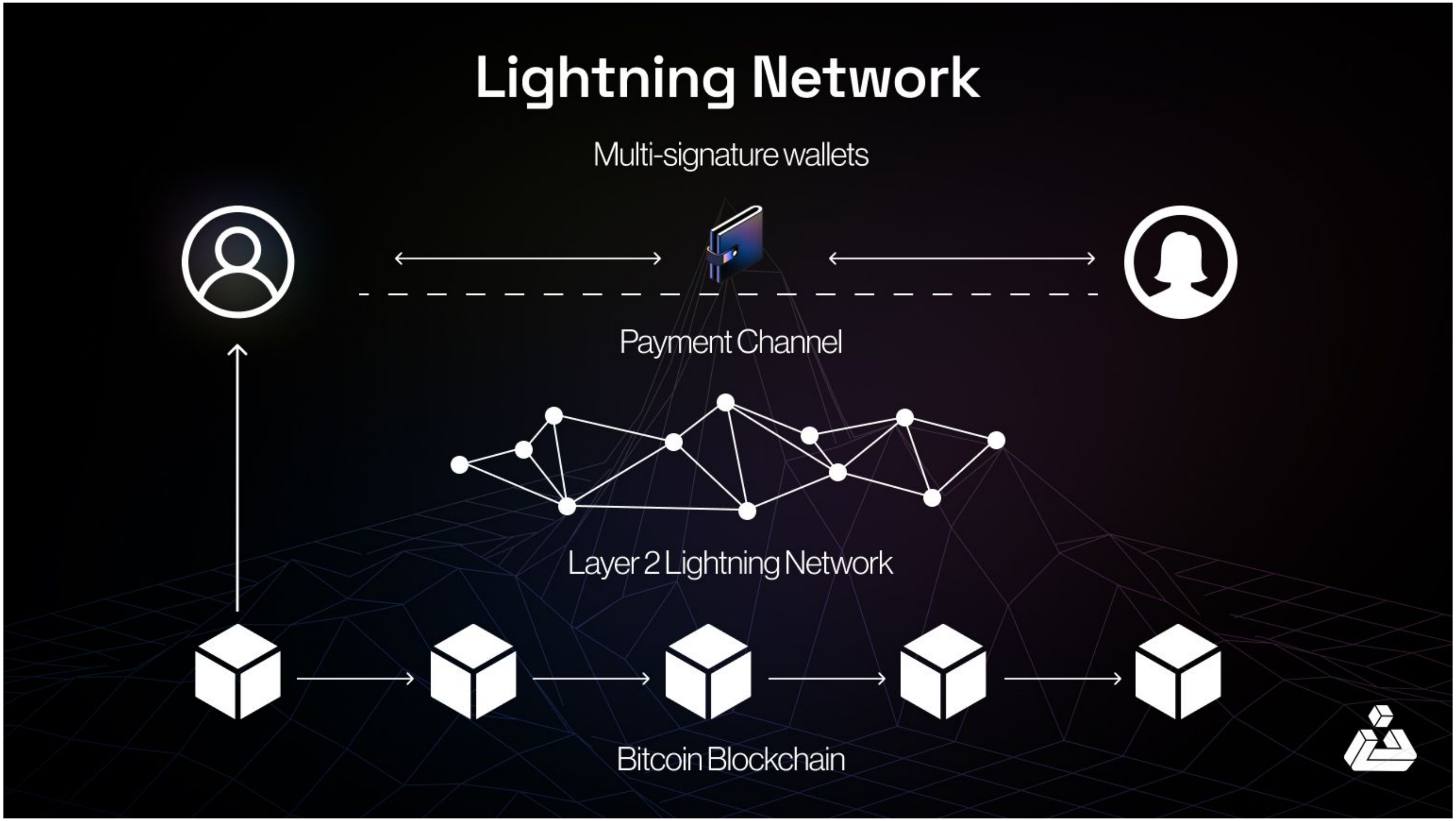
# Optimistic Rollups Blockchain Protocol

- The Optimistic Rollup is a layer 2 solution that allows for the execution of contracts on the Ethereum main network, without the need for the entire network to process each transaction.
- This can significantly improve the performance and scalability of the Ethereum network, and it allows for more complex and sophisticated dApps to be built on top of the Ethereum main network.

# Lightning Network

- Payment channels
- Fast transactions
- Low fees

# Lightning Network



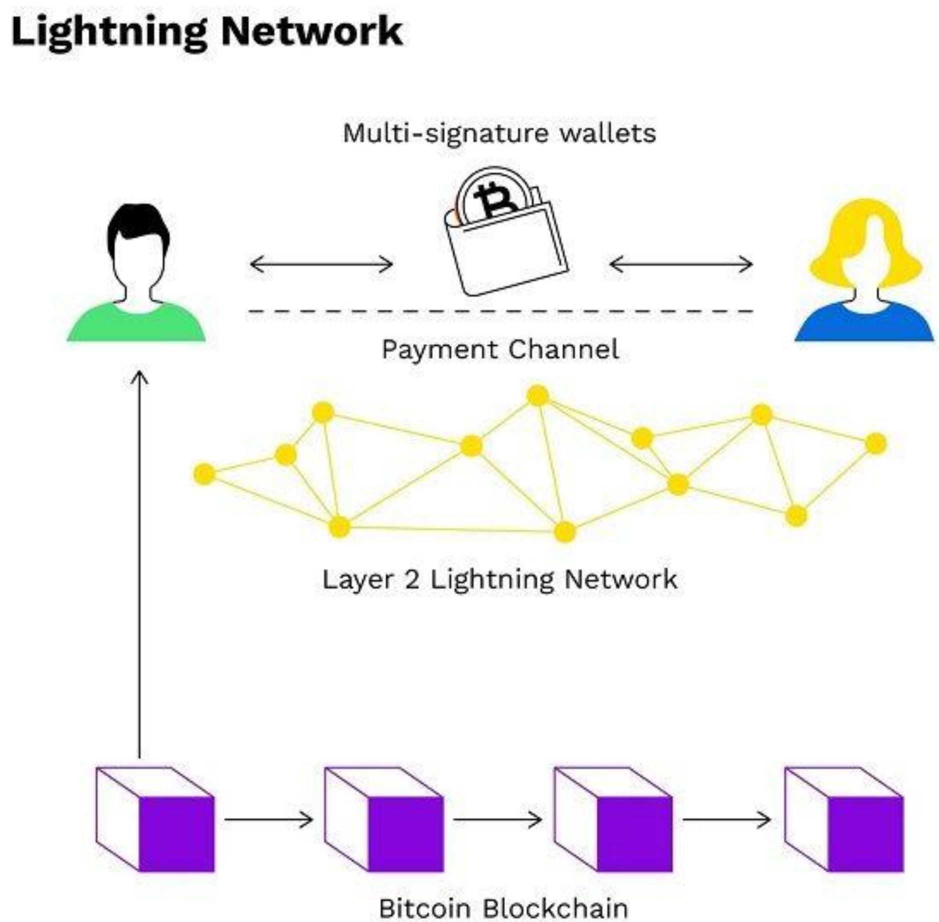
# The Lightning Network (LN)

- A Layer 2 payment protocol built on the Bitcoin blockchain to enable fast, low-cost transactions by processing them off-chain.
- By using payment channels, LN addresses scalability issues, recording only the opening and closing balances on the blockchain.

# lightning network

- Bitcoin has a layer 2 called the lightning network. Transactions performed on the lightning network are lighter, which can also be thought of as a lighting network because it makes the entire transaction set lighter on state channels.

# lightning network



# Lightning Network -off-chain transaction system

- Lightning Network consists of an off-chain transaction system built on the Bitcoin Blockchain. This system operates at the peer-to-peer (P2P) level, its applicability is based on the principle of creating two-way payment channels through which users can perform seamless cryptocurrency transactions.
- Once the two parties have agreed to create a payment gateway, they can transfer money back and forth among others wallets. Although the establishment of such payment channels will involve on-chain transactions, in return all transactions performed within that channel are off-chain, and will not need consensus of the whole system.

- As a result, these transactions can be executed quickly through smart contracts, along with low costs associated with transaction speed at a much higher level.
- To set up a payment channel, two participants need to create a multi-signature wallet with a certain amount of money available. This amount can only be accessed once both parties simultaneously provide the private keys that can be 2 or more depending on the case. This is to ensure that neither parties can access the funds without the consent of all the other parties.

Linda wants to use the lightning network to transact Bitcoin with Tina. Firstly, they both set up a payment channel, using a multi-signature wallet.

At that time, payment channels will be in the nature of a smart contract, and the multi-signature wallet will be a safe deposit box to store the desired amount of money for transactions. During the lifetime of this payment channel, Linda and Tina wanted to create as many off-chain transactions as they wanted.

Immediately after each transaction, Tina and Linda als have to sign and update their own balance sheet, which is responsible for recording the amount of coins each party holds.

When the transaction is completed, the payment channel can be closed, and the final balance sheet will be declared on the blockchain. Lightning network's smart contract will ensure that each party receives their exact amount of bitcoins based on the final version balance sheet.

So actual participants would then only have to interact with Bitcoin's blockchain network twice. Once to open the payment channel, the next time to close it. This means that all other transactions arising in the payment channel will not be performed directly on the main chain.

This is how lightning network works on the blockchain system, and we can see that users' transactions will be ensured.

# Polygon

- Ethereum compatible
- Low gas fees
- High throughput

# Polygon (POL)

- A Layer 2 blockchain protocol built on Ethereum, designed to enhance scalability, reduce transaction costs, and enable the development of decentralized applications (dApps) for Web3.
- Formerly known as **MATIC**, the native token POL powers governance, security, and transaction fees within the Polygon ecosystem.
- As a key part of the Ethereum scaling landscape, Polygon aims to build the “Value Layer of the Internet.”

# How Polygon Works

## Modified Proof-of-Stake (PoS)

- validators stake POL tokens in return for the ability to confirm transactions and earn rewards.

This consensus mechanism allows:

- – Faster block confirmation times
- – Lower energy consumption
- – Improved scalability compared to Ethereum mainnet

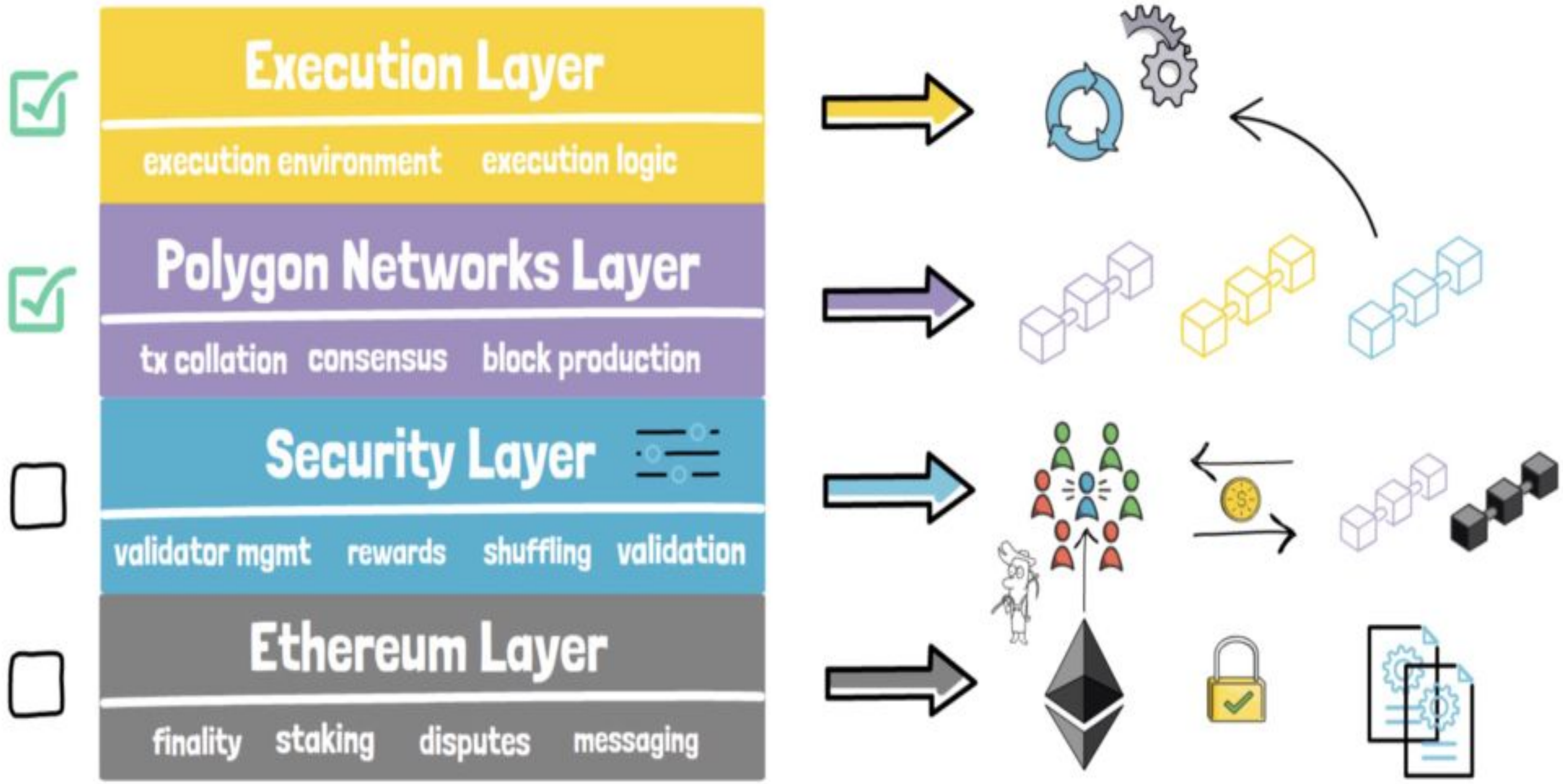
# Benefits of Using Polygon

- **Faster and Cheaper Transactions**
  - Polygon completes transaction finality in a single block, enabling **>32 transactions per second** and fees typically under **$0.01**.
- **Ethereum Compatibility**
  - Polygon is fully **Ethereum-compatible**, enabling cross-chain communication and supporting Ethereum-based smart contracts.
- **Interoperability and Developer Support**
  - Polygon provides development tools and custom blockchain deployment for projects seeking **high scalability** and **Ethereum interoperability**.

# Limitations of Polygon

- **Not a standalone blockchain–** Polygon relies on Ethereum for core security.
- **POL's limited consumer utility–** It's primarily used for staking, governance, and transaction fees rather than everyday payments.
- **Still evolving–** Like many Web3 platforms, Polygon is actively under development.

# ARCHITECTURE



# components that make up Polygon’s digital network

- Security Layer – it provides digital protection to all Polygon-compatible platforms through the ‘validator-as-a-service’ protocol. Notably, it enables the projects to gain smooth access to Ethereum’s validator pool.
- Execution Layer – it can be considered to be the run-time environment where all the blockchains that are participating can be deployed. For instance, the Ethereum Virtual Machine (EVM) is implemented and used within the layer for the execution of different smart contracts.

- Ethereum Layer – this layer helps in facilitating many internal operations like dispute resolution, checkpointing, finality, staking, and messaging that exists between all of the participating Polygon chains and Ethereum.
- Polygon Network Layer – this is described as the substrate where all of the Ethereum-compatible blockchains are deployed. It is important since it supports the smooth functioning of all the major operational processes that are happening within the network.

# Decentralized Finance

- DEX
- Lending
- Yield farming

# DeFi Ecosystem

- DEX
- Lending
- Stablecoins

# NFT Overview

- Unique tokens
- Ownership
- Digital assets

# NFT Lifecycle

- Create
- Mint
- Sell

# Summary

- Layer 2
- DeFi
- NFTs